GEOPOLITICS OF THE GLOBAL FISHERIES INDUSTRY

Supply-Chain Vulnerabilities, Cold-Chain Technology, and India's Strategic Maritime Security

A policy report with a focus on Maharashtra's Konkan Coast

Five operational modules | Strategic risk assessment | PMMSY implementation roadmap

Prepared as an independent analytical policy report for decision-support purposes

Research cut-off: 9 October 2026

Scope note: This report is not an official Government of Maharashtra, Indian Navy, or Indian Coast Guard publication. It does not imply government endorsement, access to classified information, or a formal institutional partnership.

Executive Summary

Fish is simultaneously a food-security commodity, a globally traded biological product, a source of coastal livelihoods, and—in selected contested maritime theatres—a strategic instrument. The fisheries sector's security exposure is not limited to illegal fishing. It includes opaque ownership, transshipment, labour abuse, port-state control gaps, refrigeration failures, shipping chokepoints, trade-compliance failures, and the use of civilian-appearing vessels to support state objectives. These risks intersect in the Konkan coast, where traditional fishing communities depend on seasonal marine resources while processors and exporters must meet increasingly demanding traceability and food-safety requirements.

The Food and Agriculture Organization's State of World Fisheries and Aquaculture 2024 reports that global fisheries and aquaculture production reached 223.2 million tonnes in 2022, including 185.4 million tonnes of aquatic animals and 37.8 million tonnes of algae. Aquaculture surpassed capture fisheries in aquatic-animal production for the first time. This scale makes fisheries a strategic food system, but aggregate growth masks unequal access, ecosystem stress, and vulnerabilities in logistics and governance [1].

The central recommendation is to treat Maharashtra's fisheries as an integrated system: safe and legal harvest; rapid chilling; auditable landing and handling; efficient cold-chain infrastructure; verifiable catch and health documentation; resilient export logistics; and a community-based maritime awareness network operating under clear safeguards. PMMSY can support infrastructure and livelihood objectives, but eligibility, subsidy rates, project caps, and budget availability must be checked against current operational guidelines and sanctioned state plans rather than assumed.

Key findings

• Fishery value chains are vulnerable where catch identity is lost during transshipment, mixing, reprocessing, or relabelling. Digital traceability must connect vessel, trip, landing, lot, processor, cold store, and export consignment.

• Chokepoint disruption usually affects fisheries through longer transit times, freight and insurance costs, equipment and packaging delays, fuel prices, and container availability—not necessarily through a direct shortage of fish.

• Temperature control is a food-safety control, not simply a quality upgrade. Histamine in susceptible species can form before freezing; freezing does not reliably reverse histamine already formed.

• EU access depends on country and establishment eligibility, official certification, border controls, and applicable catch-certification and hygiene rules. US imports are subject to seafood HACCP requirements and other applicable FDA controls. A cold store alone does not confer export eligibility.

• The five-district Konkan belt is not a homogeneous fishery. Species mix, landing infrastructure, market access, urban demand, seasonality, and exposure to cyclones or monsoon fishing restrictions differ substantially.

• China's maritime militia is documented in open-source research, especially in the South China Sea. However, claims that any particular fishing vessel carries covert SIGINT or AIS-spoofing equipment require vessel-specific evidence and should not be treated as a default characteristic of Chinese fishing fleets.

• Traditional fishermen can contribute to maritime domain awareness through lawful, voluntary, trained reporting channels, but they should not be tasked as covert agents or placed in situations that expose them to retaliation.

Method, Definitions, and Evidence Standard

This report combines official policy and regulatory sources, international-organisation publications, open-source maritime analysis, and strategic assessment. Numerical claims are dated where possible. Where current district-level data are not available in a harmonised public series, the report provides qualitative profiles and explicitly recommends data collection rather than inventing tonnage, prices, or export volumes.

EEZ: Exclusive Economic Zone, generally extending up to 200 nautical miles from baselines under the United Nations Convention on the Law of the Sea (UNCLOS), subject to delimitation and other legal conditions. The coastal state has sovereign rights over natural resources and specified jurisdiction, but the EEZ is not territorial sea and does not confer complete sovereignty over navigation.

DWF: Distant-water fishing, usually referring to fishing beyond a fleet's home waters, often under access agreements or on the high seas. **IUU fishing:** illegal, unreported, and unregulated fishing. **Transshipment:** transfer of catch from one vessel to another, commonly to a carrier or reefer vessel. **Cold chain:** temperature-managed handling and storage from harvest through distribution.

Evidence categories used here: (1) verified rule or published finding; (2) open-source reported behaviour; (3) analytical inference; and (4) recommended policy option. Risk rankings are qualitative and should be recalibrated using local incident, landing, temperature, and trade data.

Module 1 — The Geopolitics of Fish and ‘Fisherman Diplomacy’

1.1 Seafood as a strategic resource

Fish supports nutrition, employment, export earnings, and coastal social stability. It can also become a geopolitical instrument when states subsidise distant-water fleets, negotiate access to foreign waters, contest maritime boundaries, or use fishing activity to normalise presence in disputed areas. The strategic significance of a fishing fleet depends on its scale, state support, legal access, vessel behaviour, and relationship with maritime law-enforcement or military institutions.

The legal baseline matters. Under UNCLOS, a coastal state has sovereign rights to explore, exploit, conserve, and manage living resources in its EEZ. Other states retain navigation and overflight freedoms and other lawful uses of the sea. A large fishing fleet does not, by itself, establish a legal claim to an EEZ or change maritime boundaries. Repeated presence may, however, be used to signal resolve, pressure another state’s enforcement capacity, or create operational facts that complicate crisis management.

‘Fisherman diplomacy’ can be benign: access agreements, port calls, scientific cooperation, joint stock assessments, and fisheries development partnerships. It becomes coercive when commercial activity is used to crowd out competitors, intimidate local fishers, evade enforcement, or support a state’s position in a maritime dispute. Policy should assess conduct and evidence, not nationality alone.

1.2 Distant-water fishing: economics and regulatory grey areas

DWF economics are shaped by vessel capital costs, fuel, crew, refrigeration, catch value, access fees, subsidies, and the ability to operate far from home ports. Large vessels can remain at sea longer by using onboard freezing and at-sea transfers to refrigerated carriers. This reduces the need for frequent port returns but can make independent inspection, crew welfare checks, and catch reconciliation more difficult.

Subsidies can support fleet renewal, fuel, infrastructure, insurance, or access to finance. Subsidies are not inherently illegal; their effects depend on design and international obligations. Capacity-enhancing support can nevertheless contribute to overcapacity where management is weak. The WTO Agreement on Fisheries Subsidies, which entered into force in 2025, creates disciplines concerning certain subsidies linked to IUU fishing, overfished stocks, and fishing on the unregulated high seas. National implementation and the precise facts of each subsidy matter.

Flags of convenience or open registries can complicate oversight when beneficial ownership, operational control, crew nationality, vessel identity, and flag-state enforcement are fragmented. A flag is a legal connection, not proof of misconduct. The practical risk is a governance gap: a vessel may be owned through layered companies, chartered across jurisdictions, licensed in one state, operated by another, and land catch in a third.

AIS is a collision-avoidance and vessel-identification system, not a complete truth system. AIS data may be absent because a vessel is not required to transmit in a given circumstance, equipment is faulty, coverage is limited, or transmission is intentionally disabled or manipulated. A suspicious track is a cue for verification, not conclusive proof of illegal fishing. Analysts should compare AIS with satellite imagery, radar, vessel registries, fishing authorisations, port calls, logbooks, catch documentation, and observed fishing behaviour.

1.3 Transshipment and the traceability gap

FAO’s work on transshipment warns that inadequate regulation, monitoring, and control can create loopholes through which IUU catch enters legal supply chains [2]. Transfers at sea may be legitimate and economically efficient, but risk rises when vessels operate outside authorisation, records do not reconcile, or catch from multiple sources is mixed. A credible control regime requires advance notification, authorised transfer locations or conditions, vessel identity verification, independent observers or electronic monitoring where proportionate, and reconciliation of quantities transferred and landed.

A seafood traceability record should preserve: vessel identifier and flag; licence and fishing area; trip and catch dates; species and scientific name; quantity and product form; transshipment details; landing port; processor lot; cold-store movements; health and catch certificates; and export consignment identifier. The system must preserve corrections and audit trails. Blockchain may make later alteration more visible, but it cannot prove that a false entry was true when first submitted. Verification at source remains essential.

1.4 Comparative matrix of major actors

Actor	Strategic pattern	Typical strengths	Governance and risk questions	Implications for India
China	Large distant-water fleet; state-supported fisheries infrastructure; strong presence in parts of the Pacific and contested South China Sea; documented maritime militia activity.	Scale, processing capacity, logistics, shipbuilding, market access and state coordination.	Subsidy transparency, ownership opacity, transshipment, pressure on stocks, militia-linked conduct in specific theatres.	Use evidence-led monitoring, regional cooperation, port-state controls and catch documentation; avoid treating all Chinese vessels as militia.
Vietnam	Export-oriented capture and aquaculture sector; fisheries access and enforcement are shaped by South China Sea disputes and EU market requirements.	Processing/export capability, regional supply networks and experience with international buyers.	IUU compliance, traceability, coastal stock pressure and tensions over maritime access.	Trade compliance and fisheries diplomacy can influence investment and market access.
Indonesia	Archipelagic maritime state emphasising sovereignty, domestic value capture and enforcement against illegal fishing.	Extensive maritime jurisdiction, major fisheries resources, strong political salience of sovereignty.	Balancing enforcement with sustainable management, livelihoods and foreign investment.	Shows the importance of coast-guard capacity, licensing, landing controls and community legitimacy.
Japan	Advanced fleet, cold-chain, auction and processing systems; strong seafood demand; distant-water operations constrained by treaties and stock-management regimes.	Quality systems, technology, high-value markets, scientific management capacity.	Ageing fleet and workforce, resource sustainability, import dependence for some species.	Technology partnerships may be valuable in refrigeration, traceability, quality control and stock science.
Spain	Major EU fishing and processing actor with global operations, including distant-water activity through companies and agreements.	Industrial vessels, processing, logistics, EU regulatory expertise and established markets.	Compliance with EU fisheries rules, labour standards, subsidy rules and third-country access arrangements.	Demonstrates how regulatory systems, catch certification and market access can shape commercial conduct.

This comparison is intentionally strategic rather than a ranking of legality. Each actor contains compliant and non-compliant operators. Fleet scale, access arrangements, and state policy differ by fishery and period; the matrix should be updated with current fleet and trade data before use in formal negotiations.

1.5 Policy conclusions

• Prioritise risk-based vessel screening: flag, ownership, licence, recent track anomalies, transshipment history, and prior enforcement findings.

• Require digital catch documentation and quantity reconciliation for high-risk species and export lots.

• Support regional information-sharing through legally authorised channels and recognised fisheries-management organisations.

• Use diplomatic protest, port-state measures, sanctions where legally available, and prosecution based on evidence rather than public attribution based on nationality.

• Integrate labour standards and crew welfare into IUU risk assessment; forced labour and unsafe working conditions can coexist with fisheries crime.

Module 2 — Global Supply-Chain Networks and Strategic Chokepoints

2.1 End-to-end industrial supply chain

An industrial seafood chain can begin in coastal or distant waters and end in a supermarket, restaurant, food-service distributor, or industrial ingredient plant. A typical flow is: (1) harvest and onboard sorting; (2) chilling, icing, freezing, or processing at sea; (3) optional transshipment to a reefer vessel; (4) landing at a port or delivery to a processing hub; (5) inspection, grading, filleting, cooking, freezing, glazing, or repacking; (6) cold storage; (7) container stuffing or reefer-vessel loading; (8) customs, sanitary and catch-certificate checks; and (9) distribution to wholesale and retail markets.

Processing hubs in Southeast Asia can receive raw material from multiple fleets, process it into export-ready products, and ship to markets in Europe and North America. Such processing may create jobs and value, but can also obscure origin if records are weak. Country of processing is not necessarily country of catch. Importers and regulators therefore need to distinguish catch origin, processing origin, flag state, and exporting country.

The chain has multiple failure modes. At sea: unreported catch, mixed lots, fuel or refrigeration failures, and unsafe labour. At landing: delayed icing, inaccurate weights, and informal sales that bypass records. At processing: species substitution, cross-contamination, lot mixing, and undeclared additives. In transport: reefer breakdown, poor container pre-trip inspection, power interruptions, and route diversions. At the border: missing certificates, ineligible establishments, catch-certificate discrepancies, residue or microbiological failures, and documentation errors.

2.2 Strategic chokepoint risk matrix

Chokepoint	Fisheries-related exposure	Disruption pathways	Operational mitigations
Strait of Malacca	Connects Indian Ocean supply routes to East Asian ports and processing centres; high commercial traffic and regional transshipment activity.	Congestion, collision, piracy/armed robbery risk in nearby waters, security incidents, fuel-price shock or temporary closure.	Route and carrier diversification; port-call visibility; reefer monitoring; contingency inventory for packaging and spares; verified incident reporting.
Bab-el-Mandeb and Suez Canal	Critical Asia–Europe route for frozen seafood and equipment moving between Asian processors and European buyers.	Conflict-related attacks, war-risk premiums, rerouting around the Cape of Good Hope, longer transit and reduced schedule reliability.	Pre-agreed rerouting clauses; buffer time; continuous reefer telemetry; insurance review; buyer notification and shelf-life modelling.
Strait of Hormuz	Important to energy markets and Gulf trade; exposure can be indirect through fuel, insurance, port operations and regional demand.	Escalating tensions, restricted traffic, energy-price volatility, increased freight and insurance costs.	Fuel and freight sensitivity scenarios; diversified customer markets; no reliance on a single Gulf port or carrier.
South China Sea corridors	Important for East Asian processing, transshipment and container shipping; some waters are contested.	Harassment, collision, law-enforcement confrontation, temporary restrictions or risk aversion by operators.	Track official navigational advisories; verify carrier routing; diversify processing and shipment windows; preserve documentation for delays.
Cape of Good Hope alternative	Fallback route when Red Sea/Suez transit is avoided.	Longer voyage, higher fuel use, vessel-capacity absorption, reefer time and inventory cost.	Calculate species-specific shelf-life and quality risk; ensure power and temperature alarm response; revise delivery commitments.

The Red Sea crisis provides a practical illustration of route risk. UNCTAD reported in February 2024 that Suez transits had fallen sharply and that carriers were rerouting around the Cape of Good Hope; its analysis described longer journeys, higher freight rates, and potential spillovers into food prices and supply-chain reliability [3]. These figures describe the disruption period and should not be assumed to represent conditions in October 2026 without fresh carrier and port data.

2.3 How chokepoint disruption transmits into seafood prices

The first-order impact is time and freight. Rerouting can add days to transit, increase fuel consumption, require additional vessel capacity, and push reefer containers into longer rotations. The second-order impact is cost: war-risk insurance, emergency surcharges, port congestion, demurrage, and higher inventory finance. The third-order impact is biological: a frozen product remains stable only if the cold chain stays intact, while chilled products have a more limited time-temperature margin. The fourth-order impact is commercial: buyers may reject late cargo, retailers may switch suppliers, and exporters may lose seasonal sales windows.

Not all seafood responds alike. Frozen fillets are generally more tolerant of longer transport than fresh chilled fish, but freezer performance, packaging integrity, glaze, dehydration, oxidation, and repeated temperature cycling remain material. Live shellfish and fresh high-value fish may be more time-sensitive. A risk model should therefore be SKU-specific and include product form, core temperature, shelf life remaining, customer specifications, and border-control lead time.

2.4 Supply-chain continuity controls

• Map each SKU from landing site to buyer, including vessel, processor, cold store, port, carrier, transshipment points, and border-control location.

• Use dual sourcing where feasible, but do not confuse geographic diversification with legal or sustainability equivalence.

• Pre-qualify alternative carriers and ports, and test whether alternative routes preserve temperature, delivery window, and documentation continuity.

• Adopt reefer telemetry with door, power, supply-air and return-air readings, alarms, geofencing, and escalation ownership.

• Set a documented incident protocol: preserve logs, notify buyer and insurer, quarantine affected lots, assess product safety, and record release or disposal decisions.

• Model disruptions for 7-, 14-, and 30-day route shocks, including cash-flow effects and buyer penalties.

Module 3 — Cold-Storage Logistics, Biological Integrity, and PMMSY Alignment

3.1 Why fish degrades quickly

Fish muscle has high water activity, near-neutral pH in many species, abundant endogenous enzymes, and a surface environment that supports spoilage organisms. Once the fish dies, circulation and oxygen delivery cease; energy metabolism shifts, rigor mortis develops, and endogenous enzymes begin autolysis. Bacteria from skin, gills, gut, water, equipment, and handlers proliferate under permissive time-temperature conditions. Spoilage is species-dependent: marine psychrotrophic organisms can grow at refrigeration temperatures, while higher temperatures accelerate growth and metabolite production.

Three processes matter operationally. First, microbial spoilage produces volatile bases, sulphur compounds, amines, and other metabolites that create odours, slime, texture breakdown, and quality loss. Second, lipid oxidation affects fatty fish and polyunsaturated lipids, producing rancid flavours, colour changes, and nutrient loss; oxygen, light, metal ions, and temperature influence the rate. Third, histamine can form when bacteria decarboxylate histidine in susceptible species, including tuna, mackerel, mahi-mahi, sardines, anchovies and some related fish.

Histamine deserves special treatment because it is a chemical hazard, not simply a freshness defect. Time-temperature abuse can allow histamine formation; once formed, ordinary cooking, canning, or freezing does not reliably eliminate it. Sensory inspection cannot reliably detect unsafe histamine levels. Controls must prevent formation through rapid chilling, hygienic handling, temperature monitoring, validated time-temperature limits, supplier verification, and appropriate testing under the relevant hazard-control plan.

3.2 Cold-chain engineering by phase

Phase	Engineering and operating controls	Suggested performance indicator
Onboard harvest	Clean potable/clean seawater as appropriate; rapid bleeding/gutting where species and product plan require it; insulated bins; adequate flake/slurry ice; hygienic surfaces; lot and trip identity.	Time from catch to chilling; core-temperature record; ice-to-fish ratio validated for operation.
Blast or plate freezing at sea	Pre-cool product; select blast, plate, or brine system for product geometry; ensure air velocity/contact; avoid overloading; verify core freezing time; use calibrated probes.	Core temperature reaches validated frozen-storage target; freezer cycle completion and alarm records.
Landing and auction	Shaded, washable receiving areas; insulated crates; clean drainage; ice replenishment; temperature check at receipt; calibrated scales; segregation of rejected lots.	Landing-to-chill time; proportion of lots with complete traceability; receiving temperature compliance.
Dockside transfer	Insulated vehicles or covered transfer; minimise exposure; use hygienic, food-grade containers; separate raw and ready-to-eat products.	Transfer duration; surface and core temperatures; sanitation checklist completion.
Refrigerated road transport	Pre-cooled vehicle; calibrated logger; set-point matched to product; airflow gaps; doors closed; preventive maintenance; documented loading pattern.	Temperature excursion minutes; alarm response time; vehicle pre-trip inspection compliance.
Shore cold store	Insulated envelope, vapour barrier, appropriate refrigeration, racking, airflow, backup power, refrigerant leak management, alarms, stock rotation, sanitation and pest control.	Temperature uptime; energy per tonne; inventory dwell time; defrost and maintenance logs.
Export reefer container	Pre-trip inspection; correct set-point and ventilation; verified plug-in at terminal; continuous monitoring; seal and container ID recorded; contingency for power interruption.	Temperature excursion count; port dwell; plug-in continuity; seal and logger reconciliation.

For frozen seafood, a common operational target is storage at or below -18°C, subject to product, buyer, process, and applicable legal specifications. Chilled fish is commonly maintained close to melting ice, around 0–4°C depending on product and specification. These are operating reference points, not universal legal limits; validated HACCP plans and destination requirements govern each product. Temperature measurement should distinguish air readings from product core temperature

and should use calibrated equipment.

3.3 Critical controls and audit architecture

Every establishment should operate a hazard analysis and critical control point (HACCP)-based food-safety system, prerequisite hygiene programmes, supplier approval, water and ice quality controls, cleaning and sanitation schedules, pest control, staff hygiene, preventive maintenance, calibration, product traceability, recall capability, and corrective-action records. A paper temperature sheet is not sufficient if it is routinely filled after the fact or cannot be tied to a lot.

Minimum digital record set: unique lot ID; species and product form; vessel and trip; catch/landing date; receiving temperature; processing steps; batch inputs and outputs; freezer cycle; cold-store location; reefer/container ID; temperature logger file; certificate identifiers; and dispatch/recipient details. Data access should be role-based, with signed changes and an audit log. The system must allow an affected lot to be isolated quickly without blocking unrelated lots.

3.4 EU and US export compliance

For EU market access, seafood imports are subject to country and establishment eligibility, official health certification, border control, and applicable hygiene and traceability rules. The European Commission's guidance explains that fishery products for human consumption must originate from authorised countries and approved establishments where applicable, be accompanied by the correct official certificates, and pass the required border controls [4]. EU IUU rules also use catch certification to prevent imports of fishery products obtained through IUU fishing [5]. Product-specific and current rules must be verified before dispatch.

For the United States, FDA seafood HACCP regulations under 21 CFR Part 123 require covered seafood processors to conduct hazard analysis and implement controls for hazards reasonably likely to occur. Importers have obligations under the seafood HACCP framework, and other requirements may apply, including prior notice, facility registration where applicable, labelling, residue controls, and import alerts. The applicable rule depends on product and activity. 'Phytosanitary' is not the correct general term for fish safety; the relevant framework is food safety, animal-health where applicable, sanitary certification, and fisheries legality.

3.5 PMMSY alignment and Maharashtra delivery model

The Maharashtra Fisheries Department's public PMMSY scheme page lists a portfolio of sanctioned or recorded activities, including fish transportation services, ice factories/cold storage, deep-sea fishing boats, replacement boats and nets for traditional fishers, boat renovation for export, insurance registrations, and livelihood assistance. It reports project-level figures that should be checked against the latest official dashboard and administrative records before being quoted as current completion or beneficiary totals [6].

The state should align proposed cold-chain investment with current PMMSY operational guidelines, annual action plans, eligibility criteria, unit costs, beneficiary categories, and funding availability. Do not assume that a project qualifies simply because it is a cold store or that a particular subsidy percentage applies across all categories. Project approvals should include land and power readiness, operating model, expected throughput, maintenance financing, environmental permissions, market linkage, and measurable post-harvest-loss reduction.

Recommended programme architecture

Workstream	Intervention	Delivery and safeguards
Landing nodes	Ice plants, insulated auction halls, hygienic drainage, potable water, washable surfaces, digital weighing and lot registration.	Prioritise high-volume landings and underserved clusters; require operator training and maintenance contracts.
Village and cooperative nodes	Modular chill rooms, ice access, insulated boxes, shared reefer vehicles and solar-plus-backup power where viable.	Cooperative/SHG participation; transparent tariffs; service access for small-scale fishers; avoid stranded assets.
Export cluster	Pre-cooling, blast freezing, cold storage, testing capacity, packaging, reefer staging and digital traceability.	Link investment to buyer specifications, approved establishments, competent-authority oversight and audited HACCP.
Digital compliance	Temperature logger integration, vessel-trip-lot records, certificate management and recall workflow.	Data minimisation, role-based access, cybersecurity, offline fallback and independent verification.
Finance	Combine eligible public grants with state contribution, bank finance, cooperative equity and operator investment.	Current scheme rules control; conduct viability and sensitivity analysis; prohibit double funding.
Outcomes	Measure loss rate, temperature excursions, rejected lots, income retained by fishers, energy use and female employment.	Baseline before construction; third-party audits; public dashboard with privacy safeguards.

A practical sequence is to begin with an infrastructure and service audit, not a construction announcement. Record seasonal landing volumes by species, peak-hour queueing, current ice availability, time to first chilling, electricity reliability, distance to cold storage, truck availability, rejection reasons, and actual buyer demand. Rank projects by avoided loss and inclusion, then build a maintenance-funded network rather than isolated assets.

Module 4 — Konkan Belt Case Study and PMMSY Interventions

4.1 Regional context and data caveat

Maharashtra's Konkan coast combines dense urban fish markets, small-scale and mechanised fishing, estuarine and nearshore ecosystems, harbours, processing businesses, and export-oriented logistics. The five-district frame requested here—Thane, Mumbai, Raigad, Ratnagiri, and Sindhudurg—is useful for planning, but administrative boundaries and fisheries reporting categories may change across statistical series. Mumbai's urban market and landing systems are also not directly comparable with rural coastal districts.

Reliable district comparisons should use the latest Maharashtra fisheries statistical tables, harbour/landing-centre registers, species-level landing surveys, MPEDA export data, and local market price series. This report does not invent district-level catch tonnage, species shares, or ex-vessel prices where a consistent current series has not been verified. The profiles below are therefore operational and qualitative, suitable for prioritising data collection and infrastructure planning rather than representing audited yield estimates.

4.2 Species profile and seasonality

Common/local name	International / market name	Typical commercial handling concern	Planning implication
Paplet	Pomfret (species may include silver/black pomfrets)	High-value fish; identity and size grading matter; freshness strongly affects price.	Rapid icing, careful stacking, species-level lot records and premium-market handling.
Surmai	Kingfish / seer fish (species depends on local identification)	High-value chilled fish; large fish may require careful gutting and temperature control.	Large insulated boxes, rapid chilling, clean cutting surfaces and consistent grading.
Bangda	Indian mackerel	Oil-rich fish; histamine risk is material if time-temperature controls fail.	Immediate chilling, documented catch-to-ice time and risk-based histamine testing.
Bombil	Bombay duck	Highly perishable; sold fresh, dried or processed depending on market.	Separate fresh and drying pathways; hygienic drying, moisture control and protection from contamination.
Prawns/shrimp	Prawns / shrimp (multiple species)	Microbiological risks, melanosis/quality decline and residue requirements depending on production source.	Rapid chilling, hygienic peeling/processing, water quality and lot segregation.
Sardines	Sardines / oil sardines (species-specific)	Small pelagic fish; high volume and rapid quality decline when warm.	Bulk icing, short landing dwell, efficient sorting and local-market access.
Ribbonfish	Ribbonfish / cutlassfish	Export and domestic uses vary by size and market.	Grading, species identification and buyer-specific freezing specifications.
Crabs and cephalopods	Crab, squid, cuttlefish, octopus	Species-specific live, chilled or frozen handling requirements.	Separate product flows, validated handling plans and accurate labelling.

Seasonality is driven by monsoon conditions, the annual fishing-ban period applicable to the relevant fleet and jurisdiction, species migration, sea state, gear, market demand, and local ecological conditions. A district calendar should be prepared from at least five years of monthly landings and price data, then reviewed with fishers and scientists. A generic statewide calendar should not substitute for species- and harbour-specific evidence.

4.3 District-by-district operating profile

District	Indicative fisheries and market profile	Priority interventions to test
Thane	Creeks, estuarine systems, peri-urban landing and consumption networks; close links to Mumbai's wholesale and retail demand.	Ice and insulated transport at dispersed landing points; creek water-quality surveillance; digital lot aggregation and access for small operators.
Mumbai	Large urban demand, traditional landing communities, major auction/wholesale infrastructure and proximity to logistics services.	Hygiene and drainage upgrades; cold-room access; transparent auction data; safety and livelihood support for traditional Koli communities.
Raigad	Mix of coastal villages, estuarine and marine fisheries, and proximity to JNPA/JNPT logistics and industrial corridors.	Cluster cold chain; road access; landing-to-port traceability; assess industrial/port development impacts and fishers' access rights.
Ratnagiri	Major coastal fishing activity and harbour infrastructure including Mirkarwada; mechanised and traditional sectors.	Landing-harbour workflow, ice availability, pre-cooling, vessel safety, export-compliant handling and cooperative market linkage.
Sindhudurg	Southern Konkan coastal and estuarine resources; smaller settlements and tourism-linked local demand alongside commercial fisheries.	Decentralised ice/cold storage, insulated transport, community ownership models, weather alerts and livelihood diversification.

These are planning hypotheses, not substitutes for district baseline surveys. The department should publish the underlying dataset—month, landing centre, species, gear, quantity, price, fleet segment, and data-collection method—so that capital allocation can be independently evaluated.

4.4 Export logistics and port mapping

The requested hubs play different roles and should not be treated as interchangeable export gateways. Jawaharlal Nehru Port Authority (JNPA, historically commonly called JNPT) is a major container port serving broad international liner networks. It can be relevant to frozen and processed seafood moving in refrigerated containers, but an exporter's actual route depends on the nominated freight forwarder, shipping line, container availability, customs clearance, buyer contract, and port/terminal service. It is not necessarily the location where fish is landed or processed.

Sassoon Dock is an important Mumbai fish landing and market location. It should be understood primarily as a landing, auction, and local supply-chain node; any export consignment must use the appropriate licensed processing, certification, customs and port logistics chain. Mirkarwada in Ratnagiri is a significant fisheries harbour and landing node, with the potential for upgraded icing, hygiene, digital weighing, pre-cooling, and transport links. Minor ports and jetties may serve local landings or coastal transport but require specific validation before being described as established international seafood export gateways.

Node	Likely role in value chain	Evidence needed before investment or route claims
JNPA/JNPT	Containerised export gateway; access to shipping lines and reefer-container infrastructure.	Current carrier schedules, reefer plug availability, port dwell, freight rates, customs/health workflow and buyer routing.
Sassoon Dock, Mumbai	Landing, auction, wholesale and potential aggregation node; not automatically an export-certified processing facility.	Landing volume, hygiene audit, approved-establishment status, transport time to processor, cold-chain capacity and community access.
Mirkarwada, Ratnagiri	Landing and harbour node; potential cold-chain and processing cluster.	Current ice capacity, peak landings, power reliability, road transit time, processor demand and competent-authority approvals.
Other Konkan jetties/ports	Local landing, coastal movement, possible aggregation and road transfer.	Port capability, draft and handling constraints, legal permissions, export documentation capability and commercially confirmed service.

4.5 Text-based global export map

The following is a route-planning map, not a claim that direct scheduled services exist from every Konkan landing centre to each destination. Actual movement commonly combines road transport to an approved processor/packhouse, containerisation at a suitable logistics node, and liner shipping via one or more hub ports.

Origin / node	Domestic leg	International corridor (indicative)	Target markets and checks
Konkan landing centre → approved processor	Insulated vehicle to processor/cold store; preserve vessel-trip-lot identity.	Road to a suitable container terminal, potentially JNPA, depending on carrier and shipment design.	All markets: confirm approved facility, product eligibility, certificate set and importer specifications.

Origin / node	Domestic leg	International corridor (indicative)	Target markets and checks
JNPA/JNPT → East Asia	Reefer container; carrier-selected transshipment may apply.	West coast India → Indian Ocean → Strait of Malacca/Singapore or other carrier hub → East Asian port.	China, Japan, South Korea and other buyers only where product, establishment and import rules permit.
JNPA/JNPT → Middle East	Reefer container via carrier network; routing varies by destination and schedule.	Arabian Sea → Gulf of Oman / destination-specific Gulf or Arabian Sea port; Hormuz exposure depends on destination.	UAE, Saudi Arabia, Qatar, Oman and other markets; verify local importer registration and certificate rules.
JNPA/JNPT → EU	Reefer container and documented cold chain.	Arabian Sea → Bab-el-Mandeb/Red Sea → Suez Canal → Mediterranean/North European port; alternative via Cape of Good Hope when risk dictates.	EU country and establishment eligibility, health certificate, border control and IUU catch documentation.
JNPA/JNPT → North America	Reefer container; route depends on coast, carrier, and service.	Possible transshipment through Asian or Middle Eastern hubs, then onward to US/Canadian ports; actual route must be carrier-confirmed.	US FDA seafood HACCP/import controls; Canada's applicable import licensing and food-safety requirements.

For each lane, the exporter should retain the bill of lading, container and seal numbers, temperature logger, commercial invoice, packing list, certificate of origin where required, health certificate, catch certificate where applicable, species and scientific names, and processing/establishment identifiers. A route diagram should be generated from the actual carrier booking and vessel schedule for each shipment rather than using static claims of direct service.

4.6 PMMSY safeguards and traditional Koli communities

The policy test for PMMSY is not only whether assets are built; it is whether benefits reach small-scale fishers, women engaged in sorting and processing, crew, vendors, and cooperatives without increasing debt or excluding those unable to contribute capital. Seasonal fishing restrictions may protect breeding stocks but create immediate income pressure. Livelihood support and insurance can reduce vulnerability, yet coverage, eligibility, claim settlement, and adequacy should be audited rather than inferred from enrolment counts.

Priority safeguards include: simple application support in local languages; transparent beneficiary lists with appeal mechanisms; no exclusion solely because a fisher lacks digital access; timely transfer of eligible livelihood support; insurance awareness and claims assistance; safety equipment and training; women's participation in cooperatives and cold-chain management; and grievance channels for harbour redevelopment or access restrictions. Any vessel modernisation subsidy should be linked to safety, fuel efficiency, legal licensing, and sustainable fishing practice, not simply greater catching capacity.

A social-impact dashboard should measure real household outcomes: days of lost income during bans; payment timeliness; insurance claims paid and rejected; average price received by fishers; women's income and leadership; safety incidents; debt burden; and the share of fish handled through compliant cold chains. Aggregated reporting should protect personal data.

4.7 Konkan implementation priorities

• Conduct a district-by-district landing and cold-chain baseline before allocating capital.

• Prioritise ice, insulated transport, hygienic receiving areas and reliable power where these solve the greatest measured loss.

• Develop at least one export-readiness cluster with accredited testing, traceability, approved processing, and reefer staging rather than dispersing expensive equipment without operating plans.

• Tie public support to service levels: operating hours, affordability for traditional fishers, uptime, maintenance reserve and independently measured loss reduction.

• Create a community governance committee at each upgraded harbour with representation of traditional Koli fishers, women workers, auction participants, cooperatives, local authorities and operators.

Module 5 — China’s Fisherman Diplomacy, Maritime Militia, and India’s Response

5.1 What the maritime militia is—and is not

Open-source research by the Center for Strategic and International Studies and its Asia Maritime Transparency Initiative documents China’s maritime militia as a group of vessels that can appear commercially civilian but operate in coordination with Chinese maritime law enforcement or military objectives in the South China Sea. Research has identified vessels using ownership records, satellite imagery, observed patterns and Chinese-language sources; a 2021 CSIS study identified more than 120 vessels and assessed additional vessels as likely militia [7]. AMTI’s 2025 analysis of 2024 activity reported record numbers of militia vessels deployed to the South China Sea, many spending substantial time anchored at Chinese outposts [8].

The militia is not synonymous with the entire Chinese fishing fleet. Nor is every steel-hulled trawler a paramilitary platform. The strategic concern is the use of ostensibly civilian vessels to create presence, crowd or obstruct another state’s ships, support law-enforcement pressure, and complicate decisions about escalation. These activities sit in a grey zone between ordinary commercial fishing and overt military action.

5.2 Gray-zone coercion and operational effects

Militia-style operations can use numbers, persistence, proximity, and ambiguity. Vessels may assemble in large groups, occupy features, shadow other vessels, interfere with lawful activity, or remain near disputed areas while official coast-guard vessels provide a more overt enforcement layer. This can force the other state to choose between tolerating the behaviour, responding with law enforcement, or escalating to naval involvement. The presence of civilian crews and commercial appearance complicate attribution and proportional response.

The strategic value lies partly in the legal and political ambiguity. A fishing vessel is not automatically a warship, and the legal consequences of conduct depend on the vessel’s status, the location, the activity, and the applicable law. India’s response should be grounded in maritime law, evidence preservation, interagency coordination, and proportional action.

5.3 Maritime espionage and dual-use equipment: evidence discipline

Commercial vessels can carry navigation, communications, fish-finding, satellite communications, weather, and oceanographic equipment. Some sensors and data streams have dual-use value. However, the presence of satellite antennas, sonar, or AIS anomalies does not by itself prove espionage. Specific claims about concealed SIGINT arrays, deliberate AIS spoofing, or intelligence collection should be made only where supported by credible vessel-specific evidence, technical examination, imagery, or official findings.

A lawful maritime-security assessment can examine: identity inconsistencies; unexplained loitering near sensitive infrastructure; repeated rendezvous with vessels of concern; unusual equipment relative to declared activity; conflicting registry and ownership records; AIS behaviour inconsistent with observed movement; and unauthorised entry into restricted areas. These are indicators for corroboration, not stand-alone proof. Public reporting should distinguish observation from attribution to reduce false accusations and avoid endangering innocent crews.

5.4 IUU fishing and asymmetric impacts on developing coastal states

IUU fishing can erode food security, government revenue, employment, and confidence in coastal governance. In East Africa and parts of the western Indian Ocean, weak monitoring capacity, limited patrol assets, and fragmented jurisdiction can leave coastal states exposed to incursions or opaque access arrangements. In Latin America, distant-water fleets operating near productive high-seas areas can create enforcement and sustainability challenges, particularly where coastal states have limited surveillance resources. In South Asia, fishing disputes can intersect with livelihood insecurity, boundary ambiguity, and diplomatic tensions.

These effects are not exclusively attributable to China; fleets from multiple countries can participate in illegal or poorly regulated fishing. An evidence-led analysis should use vessel-specific cases, enforcement outcomes, catch data, and transparent licensing information. The development response should combine monitoring technology with regional cooperation, sustainable stock management, fair access agreements, local processing, and protections for artisanal fishers.

5.5 India’s coastal-security coordination model

India’s maritime security architecture includes the Indian Navy, Indian Coast Guard, state marine police, port authorities, fisheries departments, customs, intelligence agencies, and coastal communities. The operational challenge is to connect reports quickly while avoiding duplication, unclear command, or unsafe tasking. The Indian Coast Guard has a central role in maritime law enforcement and coastal security; the Navy contributes to broader maritime defence. State agencies manage local licensing, landing infrastructure and community interfaces within their mandates.

Layer	Proposed operating model	Safeguards and measures
Community reporting	Voluntary reporting through known fisheries officers, marine police or designated Coast Guard channels; standardised vessel description, location/time, behaviour and photo where safe.	No confrontation; no covert tasking; protect reporter identity; no reward system that incentivises false reports.
Fisheries administration	Maintain updated registered-vessel and licence records; support transponder and safety-device adoption where mandated and technically suitable.	Accessible subsidies; training and repair; alternatives for small boats; clear privacy and data-retention rules.
Marine police / Coast Guard	Triage reports, verify identity and location, fuse radar/AIS/satellite/port information, and determine lawful response under existing protocols.	Document source confidence; distinguish emergency, safety, fisheries and security incidents.
Navy and interagency coordination	Share assessed maritime-domain-awareness products through authorised channels; use established command and coordination structures.	Avoid creating a parallel command chain; classify information appropriately; maintain clear lead agency.
Exercises and feedback	Conduct periodic coastal-security exercises, including Sagar Kavach where scheduled by competent authorities; test reporting, response and post-incident review.	Use realistic but non-sensitive scenarios; provide community feedback; measure response time and false-report rate.

Traditional fishermen can be the ‘eyes and ears’ of maritime awareness because they are frequently at sea and possess local knowledge. This role must remain a community-safety partnership, not an intelligence burden imposed on vulnerable households. A report should never require a fisher to approach, photograph at close range, follow, or challenge a suspicious vessel. Reporting channels must be simple, available in local languages, and designed for intermittent connectivity.

5.6 Transponders, vessel identity and data governance

Transponders and vessel-monitoring devices can improve safety, identity verification and search-and-rescue response. The device model must fit vessel size, power supply, coverage, maintenance realities and the relevant legal mandate. A policy that mandates equipment without funding installation, subscriptions, repairs and training may produce non-compliance rather than security. Any scheme linkage with PMMSY should be verified against current eligible components and guidelines; no claim is made here that a particular device is automatically subsidised.

Data governance is essential. Collect only the information required for safety, fisheries management and lawful security functions; define which agency can access it; preserve logs; set retention periods; and provide a complaint and correction mechanism for inaccurate vessel records. Fisher data should not be repurposed for unrelated enforcement without lawful authority. Cybersecurity controls should cover device credentials, software updates, vendor access, spoofing detection and incident reporting.

5.7 Proposed operational indicators

- Median time from community report to acknowledgement and assessment.
- Percentage of registered active vessels with current identity, licence and emergency contact information.
- Transponder uptime and repair turnaround, disaggregated by vessel class and district.
- Number of verified safety or fisheries incidents resolved, with no target that encourages inflated reporting.
- Percentage of reported anomalies corroborated by independent data sources before attribution.
- Community participation, training completion, and complaints about privacy or unsafe tasking.
- Joint exercise lessons closed within an agreed time and reviewed by the lead competent authority.

Strategic Roadmap for the Government of Maharashtra

The recommended roadmap fuses fisheries productivity, export compliance, community resilience and maritime security. It is designed as a state-level implementation proposal, not an official directive. Delivery should be coordinated with the Union fisheries ministry, MPEDA, food-safety authorities, port and customs agencies, the Indian Coast Guard, Indian Navy and other competent bodies through existing legal and administrative arrangements.

Phase I — First 0–6 months: establish the baseline

• Create a cross-departmental steering group with fisheries, ports, food safety, disaster management, cooperatives and maritime-security liaison officers. Define mandates and decision rights.

• Publish a baseline for each target landing centre: monthly landings, species mix, price, ice availability, time-to-chill, cold-storage capacity, power reliability, rejected consignments, and household dependence on seasonal fishing.

• Map export chains from landing to processor to port to buyer. Identify the approved establishments, testing capacity, documentation gaps, shipping routes, and cold-chain handoffs actually used.

• Review PMMSY current operational guidelines, approved state action plans, project status, unspent balances, beneficiary criteria, and maintenance obligations. Verify the latest figures before public reporting.

• Set a community consultation and grievance protocol with Koli organisations, women workers, cooperatives, small-boat operators and auction participants.

Phase II — 6–18 months: pilot high-value interventions

• Select a small set of pilot nodes using a transparent scorecard: post-harvest loss reduction, volume, inclusion, export demand, power readiness, climate risk and operator capacity.

• Upgrade ice, insulated crates, hygienic landing surfaces, calibrated weighing, digital lot identifiers and temperature monitoring before expanding expensive storage capacity.

• Develop a compliant export cluster linked to approved processors, laboratories, official certification, reefer transport and confirmed buyer specifications.

• Introduce a minimum digital traceability record from vessel-trip to dispatch. Preserve paper/offline fallback and train operators to reconcile data rather than merely upload forms.

• Pilot community maritime-awareness reporting with clear privacy rules, acknowledgement targets, no-confrontation guidance, and referral to competent agencies.

Phase III — 18–36 months: scale and institutionalise

• Scale successful cold-chain models through cooperative, public-private and community-managed structures where operating costs and maintenance are demonstrably viable.

• Use independent audits to verify loss reduction, temperature compliance, cold-store uptime, fisher price realisation, gender inclusion, and grant additionality.

• Establish route-contingency protocols for chokepoint disruption, including reefer telemetry, carrier alternatives, insurance review, and buyer communication.

• Link vessel registration, licensing, safety equipment, transponder support where eligible, and training into a coherent service model for small-scale fishers.

• Publish an annual fisheries resilience and maritime-safety report with non-sensitive indicators, implementation status and corrective actions.

Proposed performance dashboard

Outcome	Baseline and target method	Reporting frequency
Post-harvest loss	Measure physical and value loss by species and landing centre; set targets only after a validated baseline.	Quarterly, seasonal comparison
Temperature compliance	Share of sampled lots with complete time-temperature records and no unaddressed excursions.	Monthly
Export readiness	Share of export lots with complete catch, health, processing and transport documentation; record rejections and root causes.	Quarterly
Infrastructure performance	Ice plant and cold-store uptime, cost per tonne, energy intensity and preventive-maintenance completion.	Monthly
Livelihood resilience	Timeliness of eligible support, insurance claim outcomes, fisher price realisation and women's participation.	Half-yearly
Maritime awareness	Report acknowledgement, verification time, corroboration rate, exercise lessons and privacy complaints.	Quarterly; sensitive data restricted
Environmental sustainability	Stock and habitat indicators, compliance with seasonal restrictions, gear impacts and bycatch reporting.	Annual, with scientific review

Funding and governance principles

Use PMMSY only for eligible components under the current guidelines and sanctioned state plan. Where appropriate, combine eligible public support with state funds, cooperative contributions, bank lending, operator investment and other legally compatible programmes. Avoid double counting grants, unsupported revenue assumptions, and capital projects without operating budgets. Every project should have a lifecycle-cost plan covering electricity, refrigerant, repairs, staffing, calibration, sanitation, cybersecurity and replacement.

A project should not be approved solely because it can absorb a grant. It should show a credible throughput forecast, access for small-scale operators, a responsible operator, service pricing, maintenance reserves, environmental compliance, and a measurable public benefit. Disbursement can be linked to milestones such as commissioning, operator training, verified uptime and demonstrated cold-chain performance, subject to scheme rules.

Final strategic judgement

Maharashtra's advantage is not simply proximity to the Arabian Sea or access to a large container port. It is the ability to connect productive fishing communities to reliable cold chains, legal and traceable catch, compliant processing, predictable export logistics and trusted maritime institutions. A fragmented approach—cold stores without ice at landing, transponders without repair support, or digital traceability without verification—will underperform. The state should invest in a connected operating system where fish quality, legal provenance, community welfare and maritime awareness reinforce one another.

Strategic priority: protect the resource, preserve the catch, prove the provenance, protect the fisher, and maintain the route.

References and Further Reading

[1] Food and Agriculture Organization of the United Nations (FAO), "FAO Report: Global fisheries and aquaculture production reaches a new record high," 7 June 2024. https://www.fao.org/newsroom/detail/fao-report-global-fisheries-and-aquaculture-production-reaches-a-new-record-high

[2] FAO, "Transshipment — Illegal, Unreported and Unregulated (IUU) fishing," including FAO's findings on monitoring gaps and risks in seafood supply chains. https://www.fao.org/iuu-fishing/tools-and-initiatives/transshipment/en/

[3] UN Trade and Development (UNCTAD), "Unprecedented shipping disruptions raise risk to global trade, UNCTAD warns," 22 February 2024. https://unctad.org/news/unprecedented-shipping-disruptions-raise-risk-global-trade-unctad-warns

[4] European Commission, Access2Markets, "Health and consumer protection for animal and plant product," updated 1 June 2026. https://trade.ec.europa.eu/access-to-markets/en/content/health-and-consumer-protection-animal-and-plant-product

[5] European Union, Council Regulation (EC) No 1005/2008 establishing a Community system to prevent, deter and eliminate illegal, unreported and unregulated fishing, consolidated text. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:02008R1005-20240109

[6] Government of Maharashtra, Department of Fisheries, "Pradhan Mantri Fisheries Scheme / Pradhan Mantri Matsya Sampada Yojana" public scheme page. Figures displayed there should be checked against current administrative records before formal citation. https://fisheries.maharashtra.gov.in/pradhan-mantri-fisheries-scheme/

[7] Center for Strategic and International Studies (CSIS), Asia Maritime Transparency Initiative, “Pulling Back the Curtain on China’s Maritime Militia,” 18 November 2021. https://www.csis.org/analysis/pulling-back-curtain-chinas-maritime-militia

[8] CSIS Asia Maritime Transparency Initiative, “Dropping the Act: China’s Militia in 2024,” 27 February 2025. https://amti.csis.org/dropping-the-act-chinas-militia-in-2024/

[9] U.S. Food and Drug Administration, Fish and Fishery Products Hazards and Controls Guidance and seafood HACCP resources. https://www.fda.gov/food/seafood-guidance-documents-regulatory-information

[10] United Nations Convention on the Law of the Sea (UNCLOS), 1982, especially provisions governing the EEZ and coastal-state rights. https://www.un.org/depts/los/convention_agreements/texts/unclos/unclos_e.pdf

[11] Ministry of Fisheries, Animal Husbandry and Dairying, Government of India, PMMSY scheme and operational guideline resources. Current operational guidelines and state annual plans should be consulted for eligibility and unit costs. https://dof.gov.in/

[12] Marine Products Export Development Authority (MPEDA), export statistics and exporter guidance. https://mpeda.gov.in/

[13] Codex Alimentarius Commission, Code of Practice for Fish and Fishery Products (CAC/RCP 52-2003), for hygiene, handling and temperature management principles. https://www.fao.org/fao-who-codexalimentarius/

[14] U.S. Code of Federal Regulations, Title 21, Part 123 — Fish and Fishery Products (Seafood HACCP). https://www.ecfr.gov/current/title-21/chapter-I/subchapter-B/part-123

Limitations and Use

This document is an analytical policy report, not legal advice, an official intelligence assessment, or an official Government of Maharashtra publication. It does not contain classified information. Geopolitical and route conditions change; verify current advisories, carrier schedules, import rules, scheme guidelines, and administrative statistics before making procurement, enforcement, or investment decisions. District-level production and price claims should be drawn from the latest official datasets and local validation. Recommendations concerning maritime security must be implemented only by competent authorities under applicable law.